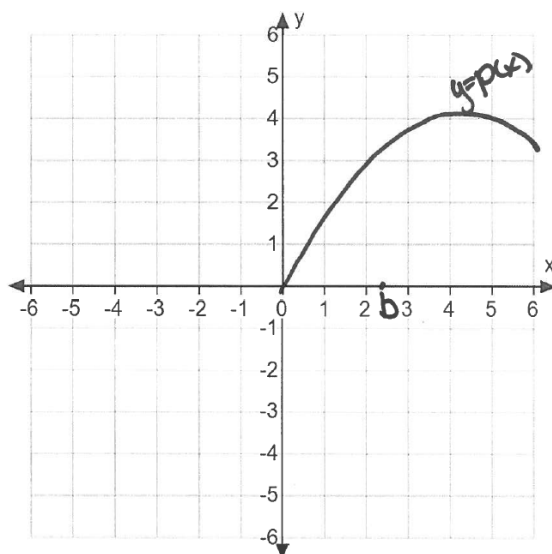


Problem 4

Problem 4

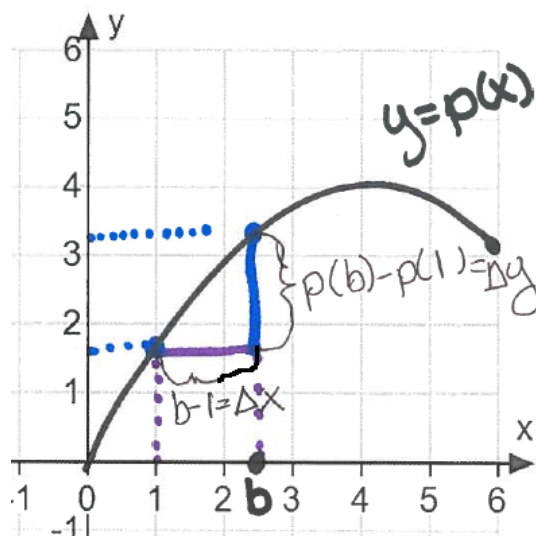
Problem 1 The graph of a function p and a point in its domain, b , are shown in the figure below.



- (a) In the figure above, draw and mark clearly the quantity $\Delta y = p(b) - p(1)$ and the quantity $\Delta x = b - 1$.

Explanation.

Problem 4



- (b) Complete the sentence. The quotient $\frac{p(b) - p(1)}{b - 1}$ is the slope...

Explanation. The quotient $\frac{p(b) - p(1)}{b - 1}$ is the slope of the secant line through the points $(b, p(b))$ and $(1, p(1))$

- (c) Complete the sentence. Provided it exists, the limit $\lim_{x \rightarrow 1} \frac{p(x) - p(1)}{x - 1}$ is the slope...

Explanation. Provided it exists, the limit $\lim_{x \rightarrow 1} \frac{p(x) - p(1)}{x - 1}$ is the slope of the tangent line to the curve $y = p(x)$ at the point $(1, p(1))$